AyuAware Onboarding Section

Technical Product Requirements Document

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1. Executive Summary

What We're Building

The Onboarding section is the user's first experience with AyuAware. Unlike traditional app onboarding with static screens and skip buttons, we're creating a **conversational introduction** where the user meets their Ayurvedic Digital Twin for the first time.

The Twin guides them through:

- Understanding what the app does (Learn → Understand → Guide)
- Setting up preferences and permissions
- A quick 5-question constitution assessment
- Receiving their initial Prakriti (constitution) insights

Why It Matters

This is where we establish the core relationship that defines the entire app experience. The user should leave onboarding feeling:

- **Understood** — The Twin already knows something about them
- **Curious** — They want to learn more about their constitution
- **Trusting** — The Twin feels like a wise, warm companion
- **Empowered** — They chose their own path forward

Key Differentiator

Most health apps treat onboarding as a hurdle to get past. We treat it as the **beginning of a relationship**. Every interaction is conversational, every question has context, and the user never feels like they're filling out a form.

2. Section Overview

2.1 The Experience Flow

```
┌──────────────────────────────────────────────────────────────────────────────────────────┐
│                                                                                          │
│   WELCOME          INTRO            PERMISSIONS      QUICK PRAKRITI      │
│   ──────────►    CAROUSEL    ──►  (Optional)  ──►                          │
│                  (3 cards)                          (5 questions)    │
│                                                                                          │
│                                          │                                               │
│                                          ▼                                               │
│                                                                                          │
│   MAIN APP    ◄──  PATH      ◄──  CONSTITUTION                                          │
│                   SELECTION        REVEAL                                               │
│                                                                                          │
└──────────────────────────────────────────────────────────────────────────────────────────┘
```

2.2 Time Expectations

Step	Expected Duration	Can Skip?
Welcome Message	5-10 seconds	No
Intro Carousel	30-60 seconds	Yes
Permissions	20-40 seconds	Yes (each one)
Quick Prakriti	60-90 seconds	No
Constitution Reveal	15-20 seconds	No

Path Selection	10-15 seconds	No
Total	**2-4 minutes**	—

2.3 Design Philosophy

Chat-First Interface

- Everything appears as messages in a chat thread
- The Twin "speaks" through message bubbles
- User responses appear as their own messages
- Interactive elements (cards, options) are embedded within the chat

Warm, Not Clinical

- No progress bars that feel like forms
- No "Step 3 of 7" anxiety
- Questions feel like genuine curiosity, not data collection
- Celebration at the end, not just "Done"

Respect User Agency

- Permissions are genuinely optional
- "Maybe later" is always accepted gracefully
- User chooses their own path at the end
- No guilt or pressure tactics

3. User Journey Breakdown

3.1 Welcome Message

What Happens: The user opens the app for the first time. After a brief splash screen, they land in a chat interface. The Twin's avatar (a glowing lotus) appears, and the first message animates in.

Twin Says:

"Namaste! I'm your Ayurvedic Digital Twin.

Think of me as a companion who learns your body's unique patterns and helps you maintain balance.

Before we begin, let me show you how we'll work together."

Design Notes:

- Avatar has a subtle pulsing glow animation (establishes it as "alive")
- Message appears with typing indicator first (feels like real-time)
- Warm, personal tone — not corporate or clinical
- Sets up the next step (carousel) naturally

3.2 Intro Carousel

What Happens: Three swipeable cards appear within a chat message, explaining the Twin's three core functions.

Card 1: "I Learn From You"

I observe patterns from your wearable devices, lab results, photos, and daily check-ins to understand your unique body.

Visual: Connected nodes showing data sources flowing together

Card 2: "I Understand You"

Using Ayurvedic wisdom, I identify your constitution (Prakriti) and current state (Vikriti) to see the complete picture of your health.

Visual: Three dosha symbols in harmony

Card 3: "I Guide You"

I provide personalized recommendations for diet, lifestyle, and daily routines that honor your individual nature.

Visual: A path/journey visualization

Interactions:

- User can swipe left/right through cards
- Progress dots show position (● ○ ○)
- "Skip Intro" text button in bottom-right
- After viewing all 3 (or skipping), flow advances automatically

Why This Matters: This isn't just feature education — it's setting expectations for an ongoing relationship. The user understands this isn't a one-time setup; the Twin will continue learning about them.

3.3 Permission Requests

What Happens: The Twin asks for two optional permissions, presented as helpful suggestions rather than demands.

Twin Says:

"To give you the most accurate guidance, I'd like to:"

Permission Card 1: Notifications

```
┌──────────────────────────────────────┐
│ 📱 Send daily check-ins              │
│                                      │
│ Best time for you?                   │
│                                      │
│ [Morning] [Afternoon] [Evening]      │
```

```
│                                          │
│                [Skip]                    │
└──────────────────────────────────────────┘
```

Permission Card 2: Health Data

```
┌──────────────────────────────────────────┐
│ ⌚ Connect wearable devices               │
│                                          │
│ This helps me understand your sleep      │
│ and activity patterns                    │
│                                          │
│ [Set up now]     [Maybe later]           │
└──────────────────────────────────────────┘
```

If User Declines: The Twin responds gracefully:

"No problem! You can always set this up later in Settings. Let's continue..."

Key Principle: Never make the user feel bad for declining. The app works without these permissions — they just enhance the experience.

3.4 Quick Prakriti Assessment

What Happens: The Twin introduces 5 quick questions to get an initial sense of the user's constitution. This is NOT the full assessment (which is 50+ questions) — it's a preview that gives both parties something to work with.

Twin Introduction:

"Now, let's get acquainted! I have 5 quick questions that will give us both a sense of your natural constitution.

These aren't tests — there are no right answers. Just choose what feels most true for you.

Ready? 🪷"

Question Format: Each question presents three options representing Vata, Pitta, and Kapha characteristics. The options are presented as visual cards the user can tap.

The 5 Questions:

#	Category	Question
1	Body Frame	"Which best describes your natural body frame?"
2	Energy	"How would you describe your energy throughout the day?"
3	Digestion	"What is your typical digestive pattern?"
4	Sleep	"How do you typically sleep?"
5	Stress Response	"When stressed, you tend to feel:"

Example Question (Body Frame):

```
Progress: Question 1 of 5  ━━○○○○○
```

```
Twin: "Which best describes your natural body frame?"
```

```
[Slim figure illustration]

Light & Slender
Difficulty gaining weight
```

```
[Medium figure illustration]

Moderate & Athletic
Gains/loses weight evenly
```

```
[Solid figure illustration]

Solid & Sturdy
Gains weight easily
```

Interaction Flow:

1. User taps an option card
2. Card shows selected state (teal border, checkmark)
3. Brief delay (400ms) for the selection to register
4. Auto-scrolls to next question
5. User's choice appears as a user message in the chat thread
6. Repeat until all 5 answered

Why These Specific Questions: These 5 areas are reliable indicators of constitutional tendencies that even a quick assessment can capture. Body frame and digestion are especially stable traits (unlike mood, which can fluctuate).

3.5 Constitution Reveal

What Happens: After the 5th question, the Twin reveals the user's preliminary constitution with a visual chart.

Twin Says:

"Fascinating! Based on these initial insights..."

Reveal Card:

```
[Dosha Visualization]
 Circular chart showing:

   Vata: 45%
   Pitta: 35%
   Kapha: 20%

"You seem to have a strong Vata
 influence with some Pitta
 qualities"

This is just our first impression.
The more we interact, the more
accurately I'll understand your
unique blueprint.
```

Critical Framing:

- "You SEEM to have" — not "You ARE"
- "First impression" — not definitive diagnosis
- "The more we interact" — sets up ongoing relationship
- Never uses medical language or makes health claims

Visual Design:

- Animated dosha chart (circle with three colored segments)
- Colors: Vata (teal), Pitta (coral), Kapha (sage green)
- Gentle scale-up animation as it appears
- Optional: subtle celebratory particles

3.6 Path Selection

What Happens: The Twin offers three paths forward, letting the user choose how they want to continue.

Twin Says:

"Would you like to:"

Options:

```
🔍 Take complete assessment (15 min)
```

```
┌──────────────────────────────────┐
│ 💬 Start with casual conversation │
└──────────────────────────────────┘
```

```
┌──────────────────────────────────┐
│ ⏭️  Explore daily rhythms          │
└──────────────────────────────────┘
```

Where Each Path Leads:

Path	Destination	Why Offer It
Complete Assessment	Full 50-question Prakriti assessment	For users who want accuracy now
Casual	Main chat interface	For users who want to explore
Daily Rhythms	Dinacharya (daily routine) section	For users who want immediate practical value

User Agency: This moment is important. The user isn't funneled into a single path — they choose based on their preference. Some people want data; others want to chat; others want actionable tips immediately.

3.7 Completion & Celebration

What Happens: After the user selects their path, the Twin celebrates and transitions to the main app.

Twin Says:

"✨ Great! We're all set up ✨

I'm here whenever you need guidance. Just ask me anything!"

Design Notes:

- Brief confetti animation (1-2 seconds, subtle)
- Twin avatar glow intensifies momentarily
- Smooth transition to selected destination
- This message stays in chat history (user can scroll back)

4. Technical Architecture

4.1 High-Level Architecture

```
┌──────────────────────────────────────────────────────────────────────────────────────────────┐
│                                                                                                  PRESENTATION LAYER
│
```

```
│
┌──────────────
```

```
│   ┌──────────────────────┐                                  │
┌──────────────────────────────────────┐     │
│   │   Onboarding Agent   │  │      Local Utilities
│   │                      │
│   │   (Claude API)       │  │  • Dosha calculator
│   │                      │
│   │                      │  │  • Message formatting
│   │                      │
│   │  Generates dynamic   │  │  • Animation timing
│   │                      │
│   │  Twin responses      │  │
│   │                      │
│   └──────────────────────┘
└──────────────────────────────────────┘     │
└──────────────────────────────────────────────────────────────────────────────────────────┘
                                                │
                                                ▼
┌──────────────────────────────────────────────────────────────────────────────────────────┐
│                                  API LAYER
│
│   ┌──────────────────────┐                                  │
┌──────────────────────────────────────┐     │
│   │   Claude API         │  │      Backend API
│   │                      │
│   │   (Anthropic)        │  │  • Save onboarding progress
│   │                      │
│   │                      │  │  • Store constitution results
│   │                      │
│   │  Real-time message   │  │  • Track analytics events
│   │                      │
│   │  generation          │  │
│   │                      │
│   └──────────────────────┘
└──────────────────────────────────────┘     │
└──────────────────────────────────────────────────────────────────────────────────────────┘
```

4.2 Component Architecture

The onboarding UI is built as a chat interface with specialized components for different message types:

Base Chat Components:

- `ChatMessageList` — Scrollable container for all messages

- `TwinMessage` — Message bubble from the Twin (left-aligned)
- `UserMessage` — Message bubble showing user's choice (right-aligned)
- `TwinAvatar` — The lotus icon with glow animation
- `TypingIndicator` — Three animated dots showing Twin is "thinking"

Interactive Components (Embedded in Chat):

- `IntroCarousel` — Swipeable cards for Learn/Understand/Guide
- `PermissionCard` — Permission request with action buttons
- `TimeSelectionPills` — Morning/Afternoon/Evening selector
- `PrakritiQuestion` — Visual option cards for assessment
- `OptionCard` — Individual tappable option within a question
- `ConstitutionReveal` — Dosha visualization with percentages
- `PathSelectionList` — Three tappable action items

Utility Components:

- `ProgressBar` — Segmented progress for Prakriti questions
- `ConfettiOverlay` — Celebration animation at completion

4.3 State Management Approach

We use **Zustand** with **AsyncStorage persistence** to manage onboarding state. This ensures:

1. **Resume capability** — If user closes app mid-onboarding, they resume where they left off
2. **Single source of truth** — All components read from the same state
3. **Persistence** — Completed onboarding status survives app restarts
4. **Simplicity** — No complex Redux boilerplate

Key State Elements:

State	Type	Purpose
`currentStep`	enum	Which step of onboarding we're on
`messages`	array	All messages in the chat thread
`isTyping`	boolean	Shows/hides typing indicator
`permissions`	object	Status of each permission request
`prakritiAnswers`	array	User's answers to 5 questions
`initialConstitution`	object	Calculated dosha percentages
`selectedPath`	string	Which path user chose at end
`isComplete`	boolean	Has user finished onboarding

4.4 Data Flow Example

Here's how data flows when a user answers a Prakriti question:

```
1. User taps "Light & Slender" option card
```

```
                    │
                    ▼
2. OptionCard component calls onSelect(option)
                    │
                    ▼
3. Parent component calls store.answerPrakritiQuestion({
     questionId: 'body-frame',
     selectedOption: 'vata'
   })
                    │
                    ▼
4. Store updates prakritiAnswers array and
prakritiQuestionIndex
                    │
                    ▼
5. Store adds user message to messages array:
   { type: 'user', content: 'Light & Slender' }
                    │
                    ▼
6. If questionIndex < 5:
   → Show next question

   If questionIndex === 5:
   → Calculate constitution using doshaCalculator
   → Store result in initialConstitution
   → Add constitution reveal to messages
   → Advance to 'constitution-reveal' step
```

5. AI Integration & Prompting Strategy

5.1 The Role of Claude in Onboarding

Claude powers the Digital Twin's personality and responses. During onboarding, Claude:

1. **Generates contextual messages** — The welcome, transitions, and celebrations
2. **Responds to user actions** — Acknowledges permissions granted/declined
3. **Personalizes the constitution reveal** — Describes the results in natural language
4. **Maintains consistent personality** — Warm, wise, never clinical

5.2 Hybrid Approach: Static + Dynamic Content

Not every message needs to be generated by Claude in real-time. We use a **hybrid approach**:

Content Type	Source	Reasoning
Welcome message	Pre-written or Claude	Could be either — consistency vs. variety
Intro carousel text	Pre-written	Must be consistent; marketing-approved

Permission descriptions	Pre-written	Legal/compliance considerations
Permission responses	Claude	Natural acknowledgment feels better
Prakriti questions	Pre-written	Validated assessment content
Constitution reveal	Claude	Personalized interpretation
Path selection prompt	Pre-written	Simple transition
Celebration	Claude or Pre-written	Could be either

Why Hybrid?

- Faster response times for static content
- Cost efficiency (fewer API calls)
- Consistency where needed (compliance, branding)
- Personalization where it matters (constitution reveal)

5.3 System Prompt for the Onboarding Twin

The system prompt establishes the Twin's identity, boundaries, and communication style. Here's the complete prompt:

SYSTEM PROMPT: Onboarding Twin

```
You are the Ayurvedic Digital Twin for AyuAware, guiding a
new user through
their onboarding experience. Your role is to be warm, wise,
and welcoming
while introducing them to Ayurvedic concepts.

## YOUR PERSONA

Identity: The user's Ayurvedic Digital Twin (you don't have a
personal name)
Tone: Warm, curious, patient, gently wise
Style: Conversational yet knowledgeable, never clinical or
preachy
Cultural approach: Honor Ayurvedic tradition while being
accessible to modern users

## CORE PRINCIPLES

1. NEVER diagnose or prescribe — you observe and guide
2. Present Ayurveda as complementary wisdom, not medical
replacement
3. Use accessible language while introducing Sanskrit terms
naturally
```

4. Celebrate the user's uniqueness rather than categorizing rigidly
5. Maintain appropriate boundaries — you're a wellness companion, not a therapist

COMMUNICATION GUIDELINES

DO:
- Use "we" to create partnership ("Let's explore...")
- Keep messages concise for chat format (2-4 sentences typically)
- Use the lotus emoji (🪷) sparingly as your signature
- Acknowledge user choices warmly without being effusive
- Frame constitution results as "impressions" not diagnoses

DO NOT:
- Overwhelm with information in single messages
- Use medical terminology or make health claims
- Pressure users about permissions
- Be overly enthusiastic or use excessive emojis
- Make the user feel bad for any choice they make

LANGUAGE EXAMPLES

Instead of: "You ARE a Vata type"
Say: "You seem to have strong Vata qualities"

Instead of: "You need to do this for your health"
Say: "You might find this supportive for balance"

Instead of: "Your Prakriti assessment shows..."
Say: "Based on our conversation so far..."

CURRENT CONTEXT

You are in the ONBOARDING flow. This is the user's first interaction with you.
Your goals:
1. Create a warm first impression that establishes trust
2. Help them understand the Learn → Understand → Guide relationship
3. Gather initial constitution insights through gentle questioning
4. Set expectations that this is just the beginning
5. Empower them to choose their own path forward

5.4 Step-Specific Prompts

For each step that requires Claude, we provide additional context:

Welcome Message Generation:

```
The user just opened the app for the first time. Generate a
warm welcome that:
1. Greets them with "Namaste"
2. Introduces yourself as their Ayurvedic Digital Twin
3. Briefly explains your role (companion who learns patterns)
4. Transitions naturally to showing how you'll work together

Keep it to 3-4 short paragraphs. End with a transition to the
intro carousel.
```

Permission Declined Response:

```
The user declined a permission (chose "Maybe later" or
similar).
Generate a graceful response that:
1. Completely accepts their choice without any hint of guilt
2. Mentions they can enable it later in Settings
3. Moves forward positively

Keep it to 2 sentences maximum. Never re-ask or pressure.
```

Constitution Reveal:

```
The user completed the 5 quick Prakriti questions. Generate a
reveal message.

Their results:
- Vata: {vata_percentage}%
- Pitta: {pitta_percentage}%
- Kapha: {kapha_percentage}%

Your response should:
1. Open with intrigue ("Fascinating! Based on these initial
insights...")
2. Describe their dosha balance in warm, accessible terms
3. Emphasize this is a first impression that will refine over
time
4. NOT mention the exact percentages (those appear in the
visualization)

Example phrasing:
```

- "You seem to have a strong Vata influence with some Pitta qualities"
- "Your responses suggest a lovely Pitta-Kapha balance"
- "You appear to have a predominantly grounded Kapha nature"

5.5 Agentic Flow Architecture

The Onboarding Agent manages the conversation flow and decides when to call Claude:

```
┌──────────────────────────────────────────────────────────────────────────────
──────────┐
│                                ONBOARDING AGENT
│
│
│
│   ┌──────────────┐     ┌──────────────┐
┌──────────────────────┐     │
│   │   Current    │     │   Action     │     │   Claude API
│   │              │     │              │     │
│   │   Context    │──►  │   Handler    │──►  │   (if needed)
│   │              │     │              │     │
│   │              │     │              │     │
│   │ • Step       │     │ Determines:  │     │ Generates:
│   │ • History    │     │ • Use cache  │     │ • Dynamic messages
│   │ • Answers    │     │ • Call API   │     │ • Personalized
text │     │
│   │ • Results    │     │ • Next step  │     │
│   └──────────────┘     └──────────────┘
└──────────────────────┘     │
│                                │
│                                ▼
│                        ┌──────────────┐
│                        │   Response   │
│                        │   + Next     │
│                        │   Step       │
│                        └──────────────┘
│
```

└──┘

Decision Logic:

User Action	Agent Decision
App opened	Load cached welcome OR generate new
Carousel completed	Use static transition text
Permission granted	Call Claude for warm acknowledgment
Permission declined	Call Claude for graceful acceptance
Prakriti question answered	No message needed (just advance)
All questions answered	Call Claude for personalized reveal
Path selected	Use static celebration OR generate new

5.6 Response Timing & Natural Feel

To make the Twin feel alive, we add intentional delays:

Event	Delay	Purpose
Before Twin message	300-800ms	Shows typing indicator, feels like thinking
After user selection	400ms	Registers the choice before moving on
Between question transitions	500ms	Prevents jarring jumps
Before reveal card	600ms	Builds anticipation

Typing Indicator Logic:

1. User completes an action
2. Immediately show typing indicator (three dots)
3. If using cached content: wait 300-500ms, then show message
4. If calling Claude: show indicator until response arrives, then show message

6. Data Models & State Management

6.1 Core Data Types

Dosha Types:

```
DoshaType: 'vata' | 'pitta' | 'kapha'

ConstitutionType:
  'vata' | 'pitta' | 'kapha' |           (single dominant)
  'vata-pitta' | 'pitta-kapha' | 'vata-kapha' |  (dual)
```

```
  'tridoshic'                                (balanced)
```

Prakriti Assessment:

```
PrakritiQuestion:
  - id: string (e.g., 'body-frame')
  - category: string (e.g., 'Physical Structure')
  - question: string
  - options: array of 3 PrakritiOptions (one per dosha)

PrakritiOption:
  - id: string
  - doshaType: DoshaType
  - title: string (e.g., 'Light & Slender')
  - description: string (e.g., 'Difficulty gaining weight')

PrakritiAnswer:
  - questionId: string
  - selectedOption: DoshaType
  - timestamp: number
```

Constitution Result:

```
InitialConstitution:
  - vata: number (percentage, 0-100)
  - pitta: number (percentage, 0-100)
  - kapha: number (percentage, 0-100)
  - dominantDosha: DoshaType | 'dual' | 'tridoshic'
  - secondaryDosha: DoshaType | null
  - constitutionType: ConstitutionType
  - confidence: 'preliminary' | 'moderate' | 'high'
```

6.2 Onboarding State Structure

```
OnboardingState:
  # Progress
  - currentStep: OnboardingStep
  - isComplete: boolean
  - startedAt: timestamp
  - completedAt: timestamp | null

  # Intro Carousel
  - introCarouselIndex: number (0-2)
  - introCarouselSkipped: boolean

  # Permissions
  - permissions:
    - notificationsRequested: boolean
    - notificationsGranted: boolean
```

```
    - notificationTime: 'morning' | 'afternoon' | 'evening' |
null
    - healthDataRequested: boolean
    - healthDataGranted: boolean

  # Prakriti Assessment
  - prakritiQuestionIndex: number (0-5)
  - prakritiAnswers: array of PrakritiAnswer

  # Results
  - initialConstitution: InitialConstitution | null
  - selectedPath: 'full-assessment' | 'casual-conversation' |
'daily-rhythms' | null

  # Chat
  - messages: array of OnboardingMessage
  - isTyping: boolean
```

6.3 Dosha Calculation Algorithm

The quick assessment uses weighted scoring:

Question Weights:

Question	Weight	Rationale
Body Frame	1.2x	Physical structure is very stable
Energy Pattern	1.0x	Moderate reliability
Digestion	1.1x	Agni is key Ayurvedic indicator
Sleep	1.0x	Moderate reliability
Stress Response	0.9x	Mental patterns can fluctuate

Calculation Steps:

1. Sum weighted scores for each dosha
2. Convert to percentages (must sum to 100%)
3. Determine dominant dosha based on thresholds:
    - If all within 15% of each other → Tridoshic
    - If top two within 10% of each other → Dual-dosha
    - Otherwise → Single dominant

Confidence Level:

- Quick assessment (5 questions) = "Preliminary"
- Partial assessment (15-30 questions) = "Moderate"
- Full assessment (50+ questions) = "High"

7. Component Requirements

7.1 Chat Components

TwinAvatar

- Circular container with lotus icon
- Three sizes: small (28px), medium (36px), large (48px)
- Optional glow animation (2-second pulse cycle)
- Glow intensity levels: subtle, medium, strong

TwinMessage

- Left-aligned message bubble
- Background: #F8FAFB
- Border radius: 4px (top-left for "tail"), 16px (other corners)
- Max width: 85% of screen
- Avatar positioned top-left of bubble
- Timestamp below message (subtle gray)

UserMessage

- Right-aligned message bubble
- Background: #4A9B8E (primary teal)
- Text color: white
- No avatar needed

TypingIndicator

- Three animated dots in a bubble
- Wave animation (dots move up/down in sequence)
- 1.2-second animation cycle
- Same styling as TwinMessage

7.2 Interactive Components

IntroCarousel

- Horizontal scrollable container
- Card dimensions: 90% screen width, auto height
- Spacing between cards: 16px
- Progress dots below cards (8x8px circles)
- "Skip Intro" text button in card footer
- Snap-to-card scroll behavior

PermissionCard

- White card with subtle shadow
- Icon + title at top
- Description text below
- Action buttons at bottom
- Full width within message bubble

TimeSelectionPills

- Horizontal row of pill-shaped buttons
- Unselected: white background, gray border

- Selected: light teal background, teal border
- Equal width distribution

OptionCard (Prakriti)

- Full-width tappable card
- Illustration at top (80x100px)
- Title + description below
- Selection state: teal border, light teal background, checkmark badge
- Minimum touch target: 44x44px

DoshaVisualization

- Circular segmented chart (180x180px)
- Three segments colored by dosha
- Animated segment growth on appear
- Legend below or beside with percentages

ProgressBar

- Segmented style (not continuous)
- 5 segments for 5 questions
- Active: teal, Inactive: light gray
- 4px gap between segments

7.3 Animation Requirements

Animation	Duration	Easing	Trigger
Message appear	400ms	ease-out	New message added
Glow pulse	2000ms	ease-in-out	Continuous
Card press	100ms	linear	Touch down
Selection confirm	400ms	ease-out	Option selected
Progress advance	300ms	ease-in-out	Question answered
Reveal scale-up	500ms	ease-out	Constitution shown
Confetti	1500ms	linear	Onboarding complete

8. API Requirements

8.1 Backend Endpoints

Start Onboarding Session

```
POST /api/v1/onboarding/start

Request:
  - userId: string
  - deviceInfo: { platform, version, deviceId }
```

```
Response:
  - sessionId: string
  - resumeStep: OnboardingStep (if returning user)
  - existingData: { permissionsGranted, prakritiAnswers }
```

Save Permissions

```
POST /api/v1/onboarding/permissions

Request:
  - userId: string
  - sessionId: string
  - permissions: {
      notifications: { granted, preferredTime },
      healthData: { granted, provider }
    }

Response:
  - success: boolean
```

Submit Quick Prakriti

```
POST /api/v1/onboarding/prakriti/quick

Request:
  - userId: string
  - sessionId: string
  - answers: array of PrakritiAnswer

Response:
  - success: boolean
  - constitution: InitialConstitution
  - description: string (AI-generated)
```

Complete Onboarding

```
POST /api/v1/onboarding/complete

Request:
  - userId: string
  - sessionId: string
  - selectedPath: string
  - metrics: { totalDurationMs, stepsCompleted,
skippedSteps }

Response:
  - success: boolean
```

```
  - userProfile: { id, initialConstitution,
onboardingComplete }
  - nextRoute: string
```

8.2 Claude API Integration

Endpoint: Anthropic Messages API (`/v1/messages`)

Model: claude-sonnet-4-20250514 (for fast responses)

Max Tokens: 500 (onboarding messages are short)

Request Pattern:

```
{
  model: "claude-sonnet-4-20250514",
  max_tokens: 500,
  system: ONBOARDING_TWIN_SYSTEM_PROMPT +
step_specific_context,
  messages: conversation_history
}
```

Error Handling:

- Timeout after 10 seconds → Use fallback cached message
- Rate limit → Use fallback cached message
- Any error → Log, use fallback, don't block user

9. Success Metrics

9.1 Completion Metrics

Metric	Target	Measurement
Onboarding completion rate	> 80%	Users who reach "complete" step / total starts
Average completion time	2-4 minutes	Timestamp at complete - timestamp at start
Drop-off by step	< 10% per step	Users who exit at each step
Path selection distribution	~33% each	Which path users choose

9.2 Engagement Metrics

Metric	Target	Measurement
Carousel engagement	> 60% view all 3	Users who swipe vs. skip
Permission grant rate (notifications)	> 60%	Granted / requested
Permission grant rate (health)	> 40%	Granted / requested

Question response time	< 5s average	Time from question shown to answer

9.3 Quality Metrics

Metric	Target	Measurement
Prakriti completion	> 95%	Users who answer all 5 questions
Return after onboarding (24h)	> 50%	Users who open app within 24h
Return after onboarding (7d)	> 30%	Users who open app within 7d
Support tickets about onboarding	< 1%	Tickets mentioning onboarding issues

9.4 Technical Metrics

Metric	Target	Measurement
Claude API response time	< 2s	Time from request to response
Fallback usage rate	< 5%	Times fallback message used
Crash rate during onboarding	< 0.1%	Crashes / sessions
State persistence success	> 99%	Successful resume after app restart

10. Implementation Considerations

10.1 Offline Handling

The onboarding should work with minimal connectivity:

Cacheable Content:

- Welcome message (pre-generate variants)
- Intro carousel (fully static)
- Permission card content (static)
- Prakriti questions (static)
- Celebration message (pre-generate variants)

Requires Connectivity:

- Personalized permission responses (can fallback)
- Constitution reveal description (can fallback to template)
- Saving progress to backend (queue and retry)

Fallback Strategy: If Claude API fails, use templated responses:

- Permission granted: "Perfect! This will help me understand you better."
- Permission declined: "No problem! You can always set this up later in Settings."
- Constitution reveal: "You seem to have a {dominant} influence with {secondary} qualities."

10.2 Accessibility Requirements

Screen Reader Support:

- All interactive elements have descriptive labels
- Progress announcements: "Question 1 of 5"
- Carousel announces: "Swipeable carousel, item 1 of 3"
- Selection confirms: "Selected Light and Slender"

Touch Targets:

- Minimum 44x44px for all tappable elements
- Preferred 48x48px for primary actions

Color Contrast:

- All text meets WCAG AA (4.5:1 ratio)
- Don't rely solely on color for information

Motion:

- Respect "reduce motion" system setting
- Provide static alternatives for animations

10.3 Analytics Events

Track these events for understanding user behavior:

Event	Properties	When
`onboarding_started`	userId, source	First message shown
`carousel_card_viewed`	cardIndex, cardId	Each card comes into view
`carousel_completed`	viewedAll, duration	After last card or skip
`carousel_skipped`	atCardIndex	Skip button pressed
`permission_requested`	permissionType	Permission card shown
`permission_granted`	permissionType, value	User grants
`permission_declined`	permissionType	User declines
`prakriti_question_shown`	questionId, questionIndex	Question appears
`prakriti_question_answered`	questionId, answer, responseTime	User selects
`constitution_revealed`	dominantDosha, constitutionType	Reveal shown
`path_selected`	pathId	User chooses path
`onboarding_completed`	totalDuration, pathSelected	Celebration shown

10.4 Error Handling

Network Errors:

- Show subtle "offline" indicator if no connection

- Queue state changes for sync when online
- Never block the user from progressing locally

API Errors:

- Claude timeout → Use fallback message
- Backend save fails → Queue for retry, continue flow
- Log all errors for monitoring

State Errors:

- Invalid state → Reset to nearest valid step
- Corrupted storage → Clear and restart onboarding
- Always allow user to "start over" if stuck

10.5 Testing Priorities

Unit Tests:

- Dosha calculator with various answer combinations
- State transitions for each step
- Message formatting functions

Integration Tests:

- Complete flow from start to finish
- Resume flow after app restart
- Permission grant/decline paths
- All Prakriti answer combinations

E2E Tests:

- Happy path completion
- Skip paths (carousel, permissions)
- Offline/online transitions
- Accessibility with screen reader

Manual QA:

- Timing and animations feel natural
- Personality consistency across flows
- Edge cases (rapid tapping, back button, etc.)

Appendix A: Content Reference

Twin Message Templates

Welcome:

Namaste! I'm your Ayurvedic Digital Twin.

Think of me as a companion who learns your body's unique patterns and helps you maintain balance.

Before we begin, let me show you how we'll work together.

Permission Granted (Notifications):

Wonderful! I'll check in with you during the {time_of_day}. These moments help me understand your daily rhythms.

Permission Granted (Health Data):

Great! This will help me see the full picture of your sleep, activity, and recovery patterns.

Permission Declined:

No problem! You can always set this up later in Settings. Let's continue...

Prakriti Introduction:

Now, let's get acquainted! I have 5 quick questions that will give us both a sense of your natural constitution.

These aren't tests — there are no right answers. Just choose what feels most true for you.

Ready? 🪷

Path Selection Prompt:

Would you like to:

Celebration:

✨ Great! We're all set up ✨

I'm here whenever you need guidance. Just ask me anything!

Appendix B: Dosha Reference

Vata (Air + Ether)

- **Qualities:** Light, dry, cold, mobile, subtle
- **Body:** Thin frame, difficulty gaining weight
- **Energy:** Variable, comes in bursts
- **Digestion:** Irregular, prone to gas
- **Sleep:** Light, interrupted
- **Stress:** Anxiety, worry, fear
- **Color in app:** #7AB8AC (light teal)

Pitta (Fire + Water)

- **Qualities:** Hot, sharp, light, oily, liquid
- **Body:** Medium, athletic build
- **Energy:** Strong, focused, goal-driven
- **Digestion:** Strong, cannot skip meals
- **Sleep:** Moderate, may wake with ideas
- **Stress:** Irritation, frustration, criticism
- **Color in app:** #E89B7F (warm coral)

Kapha (Earth + Water)

- **Qualities:** Heavy, slow, cool, oily, stable
- **Body:** Solid, sturdy frame
- **Energy:** Steady, slow to start, long-lasting
- **Digestion:** Slow, can skip meals easily
- **Sleep:** Deep, heavy, hard to wake
- **Stress:** Withdrawal, sadness, feeling stuck
- **Color in app:** #A3B899 (earthy sage)

End of Document